

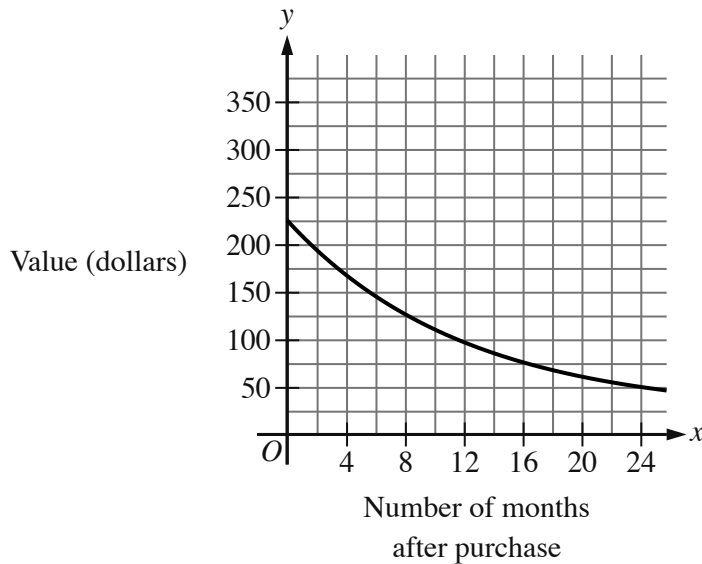
☒ EASY

☐ ADVANCED MATH

Advanced Math

10 questions · ~15 min

14

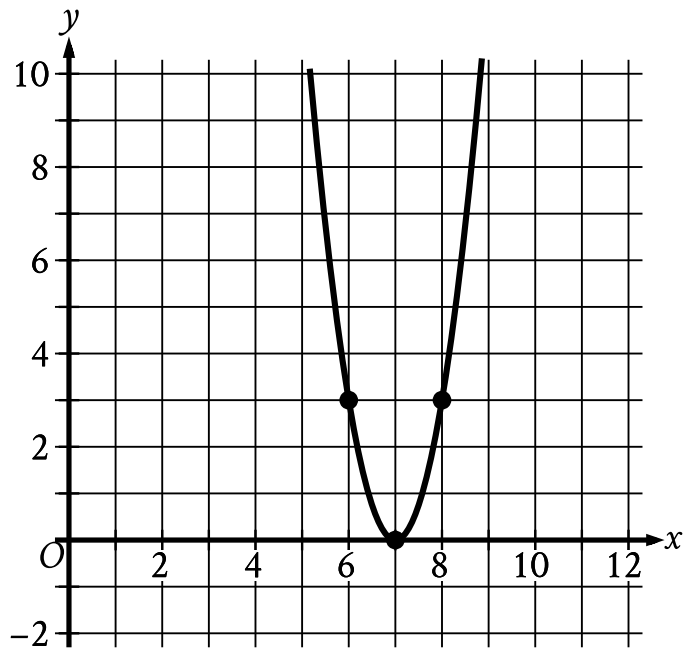


- Moving from left to right:
 - The curve is in quadrant 1.
 - The curve trends down gradually.
- The curve passes through the following approximate points:
 - (0 comma 225)
 - (24 comma 50)

The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

- A. The estimated value of the tablet was \$ 225 when it was purchased.
- B. The estimated value of the tablet 24 months after it was purchased was \$ 225.
- C. The estimated value of the tablet had decreased by \$ 225 in the 24 months after it was purchased.

- D. The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.



- The parabola opens upward.
- The vertex is at point (7 comma 0).
- The parabola passes through the following points:
 - (6 comma 3)
 - (7 comma 0)
 - (8 comma 3)

The x -intercept of the graph shown is $x, 0$. What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)
(3)	(3)	(3)	(3)
(4)	(4)	(4)	(4)

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

The function f is defined by $fx = 8\sqrt{x}$. For what value of x does $fx = 48$?

- A. 6
- B. 8
- C. 36
- D. 64

Q4

GRID-IN ADVANCED MATH

The function f is defined by $fx = x^2 + x + 71$. What is the value of $f2$?

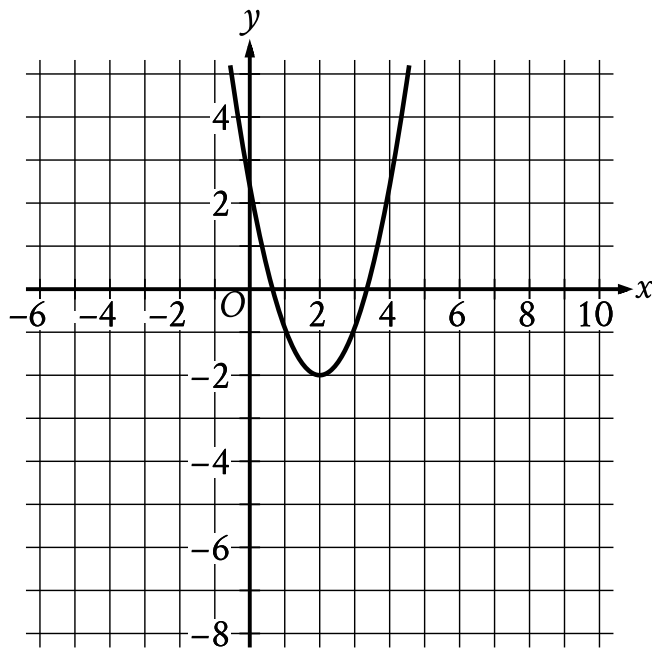
STUDENT-PRODUCED RESPONSE

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.	/	.	.
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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

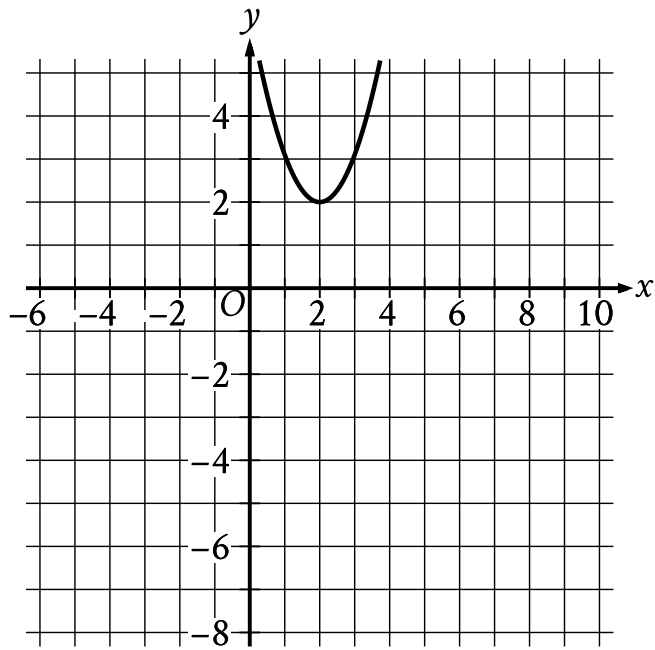
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The parabola opens upward.
- The vertex is at the point (2 comma negative 2).
- The parabola passes through the following points:
 - (1 comma negative nine tenths)
 - (2 comma negative 2)
 - (3 comma negative nine tenths)

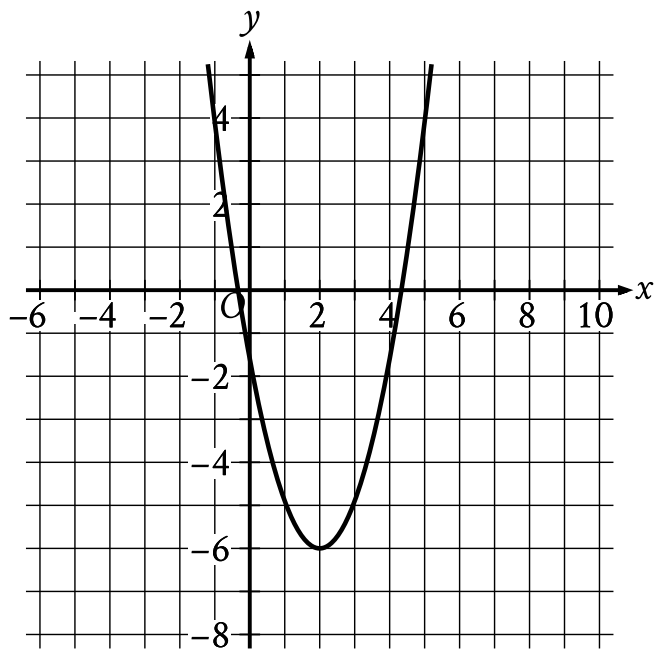
The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

A.



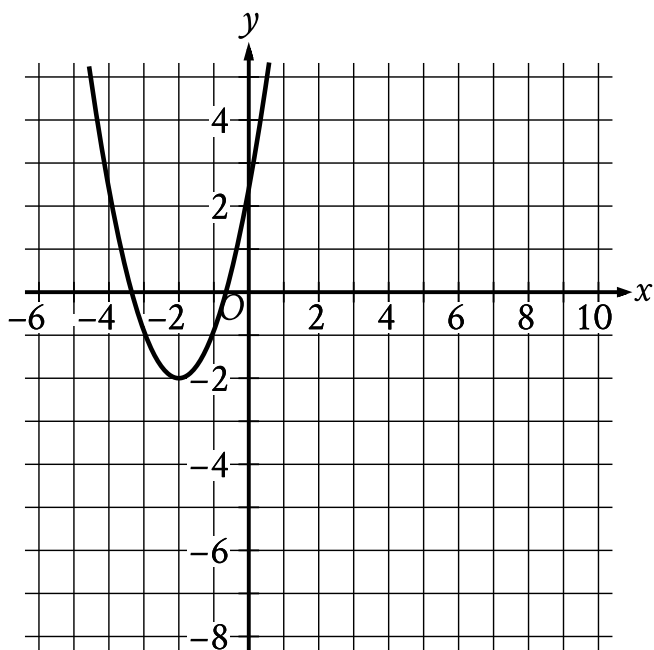
- The parabola opens upward.
- The vertex is at the point (2 comma 2).
- The parabola passes through the following points:
 - (1 comma $\frac{31}{10}$)
 - (2 comma 2)
 - (3 comma $\frac{31}{10}$)

B.



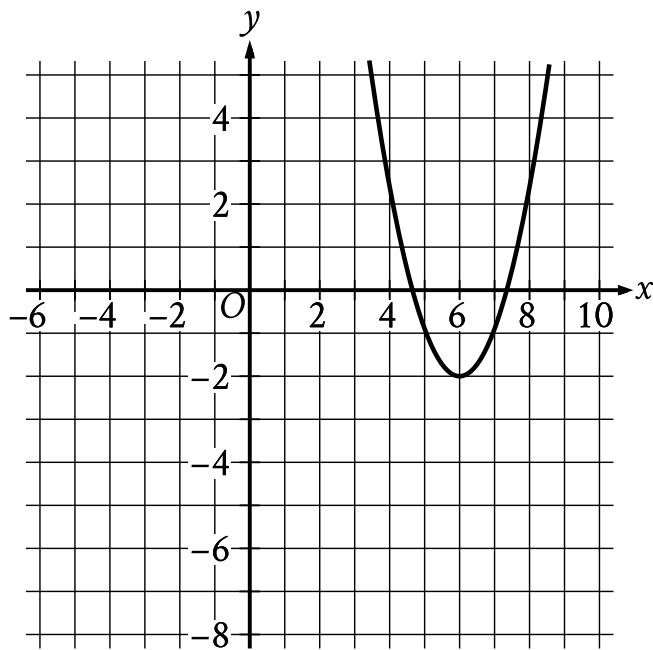
- The parabola opens upward.
- The vertex is at the point (2 comma negative 6).
- The parabola passes through the following points:
 - (1 comma negative $\frac{49}{10}$)
 - (2 comma negative 6)
 - (3 comma negative $\frac{49}{10}$)

C.



- The parabola opens upward.
- The vertex is at the point (negative 2 comma negative 2).
- The parabola passes through the following points:
 - (negative 3 comma negative nine tenths)
 - (negative 2 comma negative 2)
 - (negative 1 comma negative nine tenths)

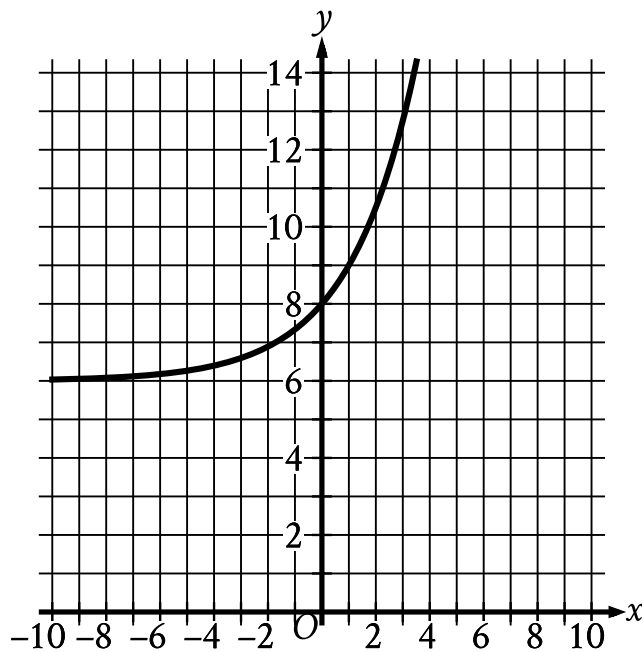
D.



- The parabola opens upward.
- The vertex is at the point (6 comma negative 2).
- The parabola passes through the following points:
 - (5 comma negative nine tenths)
 - (6 comma negative 2)
 - (7 comma negative nine tenths)

The kinetic energy, in joules, of an object with mass 9 kilograms traveling at a speed of v meters per second is given by the function K , where $Kv = \frac{9}{2}v^2$. Which of the following is the best interpretation of $K34 = 5,202$ in this context?

- A. The object traveling at 34 meters per second has a kinetic energy of 5,202 joules.
- B. The object traveling at 340 meters per second has a kinetic energy of 5,202 joules.
- C. The object traveling at 5,202 meters per second has a kinetic energy of 34 joules.
- D. The object traveling at 23,409 meters per second has a kinetic energy of 34 joules.



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point (0 comma 8).
 - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
 - (0 comma 8)
 - (1 comma 9)

What is the y-intercept of the graph shown?

- A. -8,0
- B. -6,0
- C. 0,6

D. 0,8

Q8

GRID-IN ADVANCED MATH

The y -intercept of the graph of $y = x^2 + 31$ in the xy -plane is $0, y$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function f is defined by $fx = x^3 + 15$. What is the value of $f2$?

A. 20

B. 21

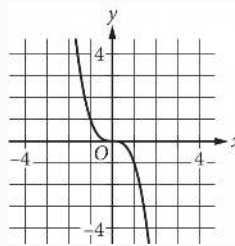
C. 23

D. 24

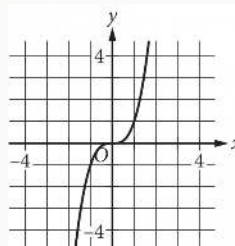
x	y
0	0
1	1
2	8
3	27

The table shown includes some values of x and their corresponding values of y . Which of the following graphs in the xy -plane could represent the relationship between x and y ?

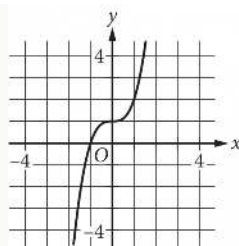
A.



B.



C.



D.

